

43.14 General

Key to 1.5/9

A	Hydraulics	1.A	= 1st axle
B	Hydraulic control pressure	2.A	= 2nd axle
C	Brake pressure	AS	= Trailer control
D	Control pressure	BV	= Actuating valve
E	Supply	DS	= Accumulator
F	Supply control pressure	MB(N)	= Engine brake (auxiliary load)
		DE	= Pressure generation
		KO	= Control switch
1.1	Air compressor	20.3	Preload cylinder
2.2	Pressure regulator	20.5	Spring-loaded parking brake
3.1	Antifreeze unit	26.5	ALB modulator
4.4	Four-circuit protection valve	28.1	2-way valve trailer operation
5.1a	Air tank circuit 1	29.1	Pressure reducing valve
5.1b	Air tank circuit 2	29.4	Pressure reducing valve, switchover
6.1	Water separator	32.1	Switchover valve, 4-wheel drive and differential lock
8.4	Non-return valve with limited return flow	33.1	Switchover valve engine brake
9.1	Warning lamp, red hydraulics	34.1a	Brake fluid reservoir circuit 1
9.1a	Warning lamp, red parking brake	34.1b	Brake fluid reservoir circuit 2
9.3	Stoplight	35.3	Coupling head brake (2-circuit)
10.1	Electrical pressure switch	35.4	Coupling head supply (2-circuit)
10.2	Electrical pressure switch	35.9	Coupling head (single circuit)
11.2	Double pressure gauge	38.2	Test connection spring-loaded cylinder
13.2	Service brake valve	38.2a	Test connection circuit 1
14.2	Parking brake valve	38.2b	Test connection circuit 2
17.1	Trailer control valve (1-line)	38.3	Test connection ALB modulator
18.5	Trailer control valve (2-line)	39.1	Vent
19	Fixed caliper		
19.3	Tandem master brake cylinder		
20.1a	Piston cylinder 4-wheel drive		
20.1b	Piston cylinder differential lock		
20.1c	Piston cylinder engine brake		

UZ42-0253

Refer to 43.11-1.5/1 to 1.5/28 for symbols.

Key to 1.5/11

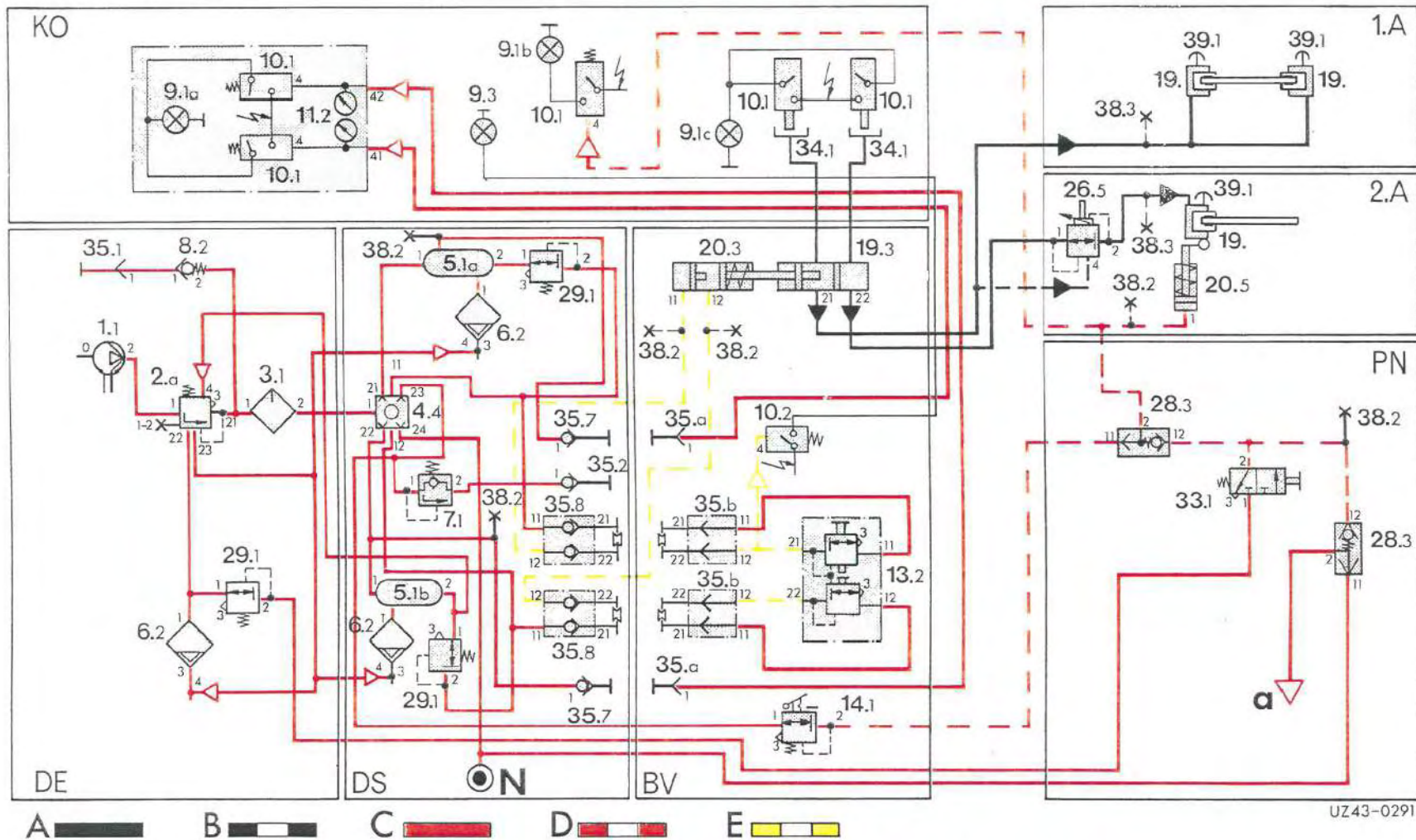
A Hydraulics
B Hydraulic control pressure
C Brake pressure
D Control pressure
E Supply
F Supply control pressure

1.A = 1st axle
2.A = 2nd axle
AS = Trailer control
BV = Actuating valve
DS = Accumulator
MB(N) = Engine brake (auxiliary load)
DE = Pressure generation
KO = Control switch

1.1 Air compressor
2.2 Pressure regulator
3.1 Antifreeze unit
4.4 Four-circuit protection valve
5.1a Air tank circuit 1
5.1b Air tank circuit 2
6.1 Water separator
8.4 Non-return valve with limited return flow
9.1 Warning lamp, red hydraulics
9.1a Warning lamp, red parking brake
9.3 Stoplight
10.1 Electrical pressure switch
10.2 Electrical pressure switch
11.2 Double pressure gauge
13.2 Service brake valve
14.2 Parking brake valve
17.1 Trailer control valve (1-line)
18.5 Trailer control valve (2-line)
19 Fixed caliper
19.3 Tandem master brake cylinder
20.1a Piston cylinder 4-wheel drive
20.1b Piston cylinder differential lock
20.1c Piston cylinder engine brake

20.3 Preload cylinder
20.5 Spring-loaded parking brake
26.5 ALB modulator
28.1 2-way valve trailer operation
29.1 Pressure reducing valve
29.4 Pressure reducing valve, switchover
32.1 Switchover valve, 4-wheel drive and differential lock
33.1 Switchover valve engine brake
34.1a Brake fluid reservoir circuit 1
34.1b Brake fluid reservoir circuit 2
35.3 Coupling head brake (2-circuit)
35.4 Coupling head supply (2-circuit)
35.9 Coupling head (single-circuit)
38.2 Test connection spring-loaded cylinder
38.2a Test connection circuit 1
38.2b Test connection circuit 2
38.3 Test connection ALB modulator
39.1 Vent

Functional Diagram (435.160,



Dual circuit hydraulic system and dual-circuit air brake system, ALB on rear axle, spring-loaded parking brake with pneumatic emergency release mechanism.

Refer to 43.11 – 1.5/1 to 1.5/28 for symbols. For auxiliary loads, see 1.6/2.

Explanations to 1.5/13

A Hydraulics
B Hydraulic control pressure
C Supply pressure
D Supply control pressure
E Brake control pressure

KO = Checking switch
DR = Pressure generation
DS = Accumulator
(N = Connection auxiliary)
BV = Actuation valve
1. A = 1st axle
2. A = 2nd axle
PN = Pneumatic emergency release

1.1 Air compressor
2a Pressure regulator
3.1 Defrosting pump
4.4 Four-circuit protection valve
5.1a Air tank
5.1b Air tank
6.2 Water separator
7.1 Overflow valve
8.2 Check valve
9.1a Control light
9.1b Control light
9.1c Control light
9.3 Stoplight

single-cylinder
cut out pressure 18-0.5 bar

opening pressure 7-0.3 bar
circuit 1 (capacity 10l)
circuit 2 (capacity 10l)
with automatic exhaustion and control connection
with backstreaming
filler line front
supply pressure
parking brake
brake fluid level

10.1 Circuit breaker
10.2 Circuit closer
11.2 Double pressure gauge
13.2 Brake valve (dual-circuit)
14.1 Parking brake valve for spring-loaded cylinder
19 Hydraulic cylinder
19.3 Hydraulic master cylinder
20.3 Pneumatic piston cylinder
20.5 Spring-loaded cylinder
26.5 Automatic brake pressure
28.3 Shuttle valve
29.1 Pressure reducing valve
33.1 ½-way valve
34.1 Brake fluid reservoir
35a Coupling head
35b Coupling head
35.1 Coupling head
35.2 Coupling head
35.7 Coupling head
35.8 Coupling head
38.2 Test connection
38.3 Test connection
39.1 Air bleed

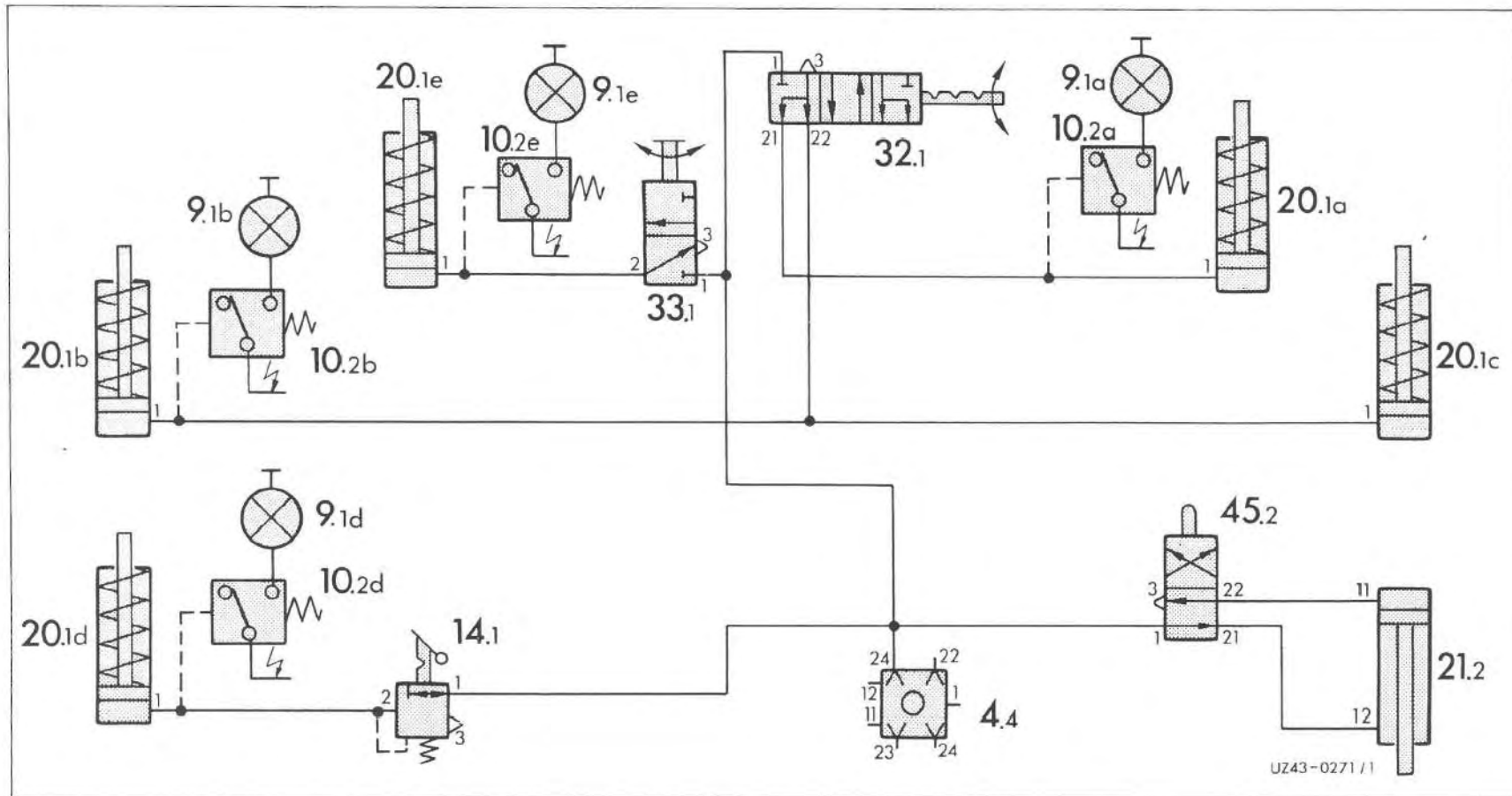
for spring-loaded cylinder

(dual-circuit)
(dual-circuit)
(pulling)
modulator (ALB)

9.5 ± 0.15 bar
pneumatic emergency release

filler line front
filler line rear
quick coupler
coupling head double
pneumatic
hydraulics

Functional Diagram Auxiliary Load (N) (435.110/111/113)



- | | | |
|--|---|--|
| 4.4 Four-circuit protection valve | 10.2d Electrical pressure switch PTO shaft | 20.1e Piston cylinder power take-off |
| 9.1a Indicator lamp, 4-wheel drive | 10.2e Electrical pressure switch power take-off | 21.2 Piston cylinder double-acting planetary gearbox train |
| 9.1b Indicator lamp, differential lock | 14.1 Switchover valve PTO shaft | 32.1 Switchover valve 4-wheel drive differential lock |
| 9.1d Indicator lamp, PTO shaft | 20.1a Piston cylinder 4-wheel drive | 33.1 Switchover valve power take-off |
| 9.1e Indicator lamp, power take-off | 20.1b Piston cylinder differential lock FA | 45.2 Switchover valve planetary gear |
| 10.2a Electrical pressure switch 4-wheel drive | 20.1c Piston cylinder differential lock RA | Refer to 43.11 – 1.5/1 to 1.5/28 for symbols |
| 10.2b Electrical pressure switch differential lock | 20.1d Piston cylinder PTO shaft | |

UZ43-0271 / 1



Explanations to 1.6/2

- a Coming from shuttle valve (28.3)
- b Coming from four-circuit protection valve, connection 24 (4.4)

A Planetary gearbox train (high and low range)

- 21.2 Cylinder
- 45.4 ½-way valve

B Exhaust valve

- 10.2 Circuit closer
- 20.5 a Spring-loaded cylinder
- 33.1 a ¾-way valve
- x Connection socket trailer (54.9)

C Fast special drive

- 9.1 a Control light
- 10.2 Circuit closer
- 20.5 b Spring-loaded cylinder
- 33.1 b ¾-way valve

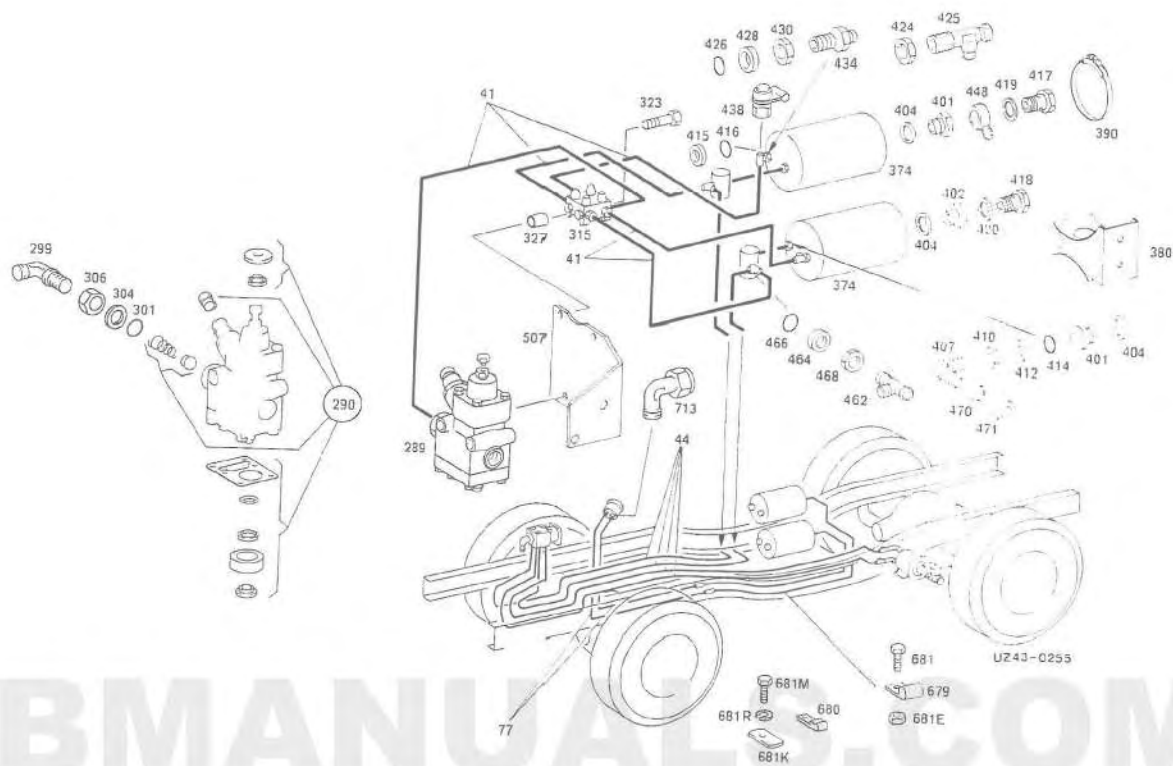
D Four-wheel drive and differential lock

- 9.1 b Control light (four-wheel drive)
- 9.1 c Control light (differential lock)
- 10.2 Circuit closer
- 20.5 c Spring-loaded cylinder (four-wheel drive)
- 20.5 d Spring-loaded cylinder (differential lock)
- 32.1 ¾-way valve

E Unit admission

- 30.1 Pressure reducing valve
- 33.1 c ¾-way valve
- 36.1 Line filter
- c Admission FA
- d Admission RA
- e Admission gearbox

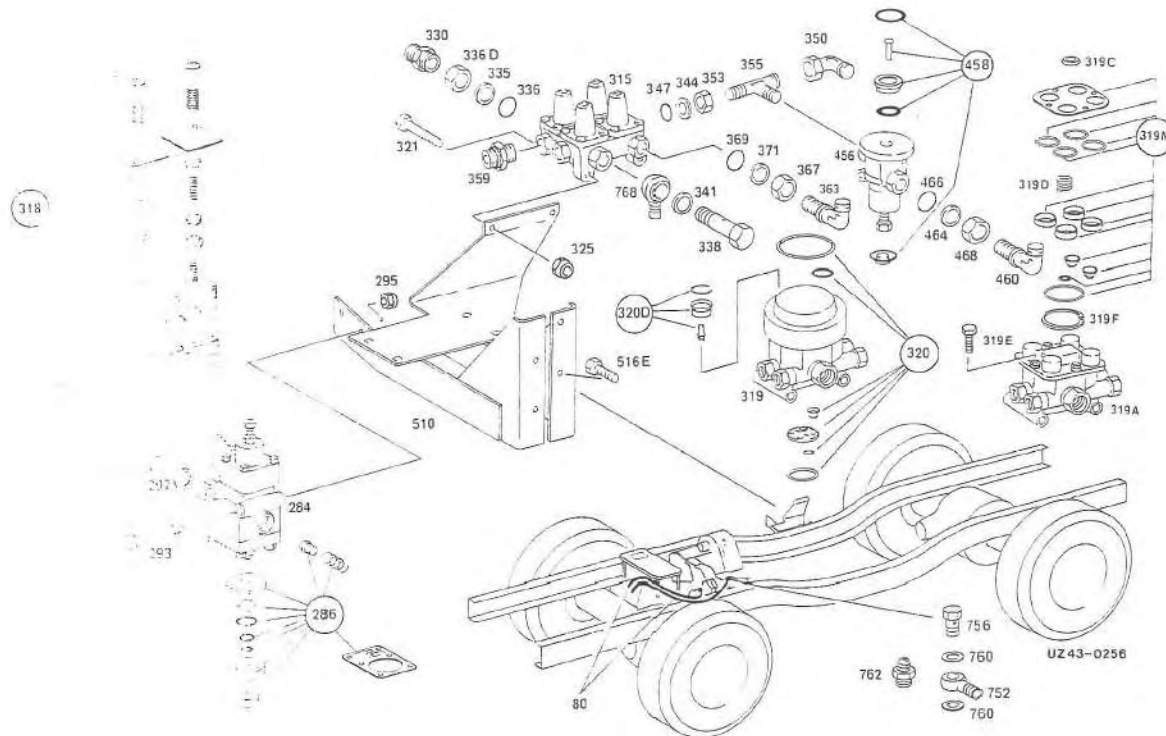
Exploded View



Brake System

41	Line	419	Sealing ring
44	Hose	420	Sealing ring
77	Hose	424	Nut
289	Pressure regulator	425	Pipe socket
290	Repair kit	426	Sealing ring
299	Pipe elbow	428	Thrust ring
301	Sealing ring	430	Nut
304	Thrust ring	434	Pipe socket
306	Nut	438	Test connection
315	Four-circuit protection valve	448	Ring piece
323	Screw	462	T-piece
327	Air tank	464	Thrust ring
380	Bracket	466	Sealing ring
390	Clip	468	Nut
401	Screw fitting	470	Thrust ring
402	Screw fitting	471	Sealing ring
403	Sealing ring	507	Bracket
407	Pipe elbow	679	Clip
410	Nut	680	Clip
412	Thrust ring	681	Bolt
414	Sealing ring	681E	Nut
415	Thrust ring	681K	Gasket
416	Sealing ring	681M	Bolt
417	Banjo bolt	681R	Lock washer
418	Banjo bolt	713	Pipe socket

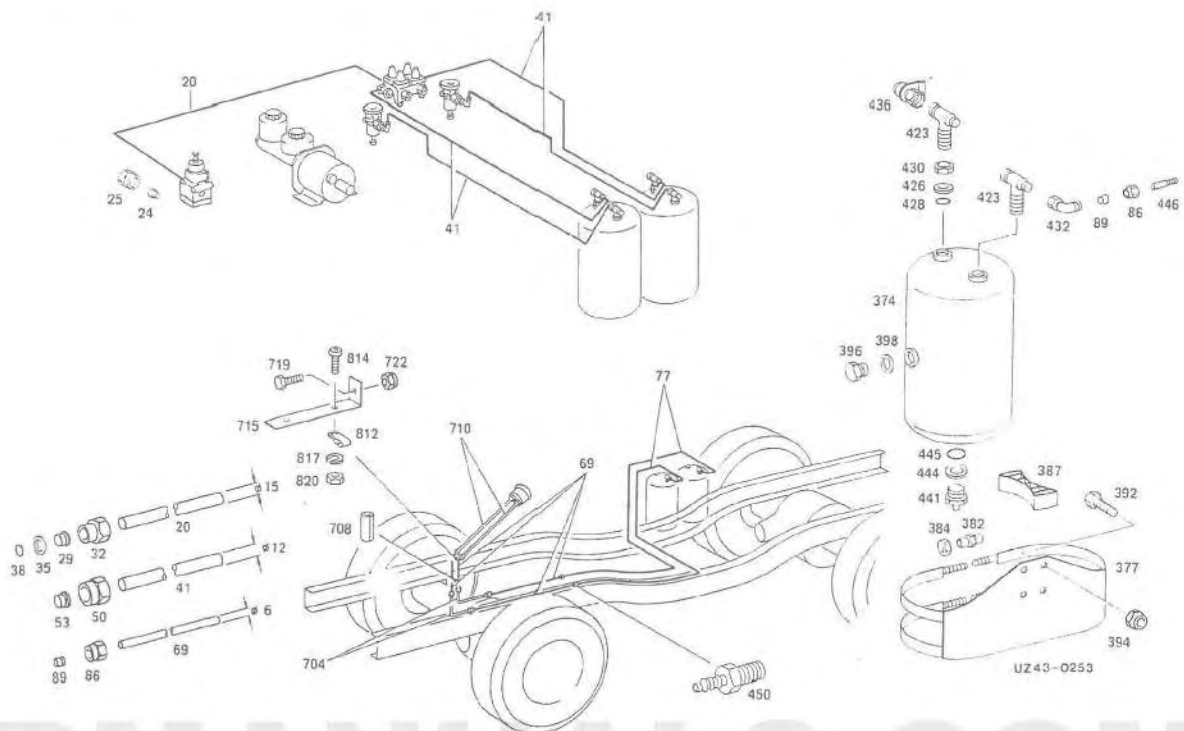
Exploded View



Brake System

80	Hose	341	Sealing ring
284	Pressure regulator	344	Thrust ring
286	Repair kit	347	Sealing ring
293	Bolt	350	Pipe socket
295	Nut	353	Nut
297	Sealing ring	355	Pipe socket
309	Double adapter	359	Threaded union
315	Four-circuit protection valve	363	Pipe socket
318	Repair kit	367	Nut
319	Four-circuit protection valve	369	Sealing ring
319A	Casing	371	Thrust ring
319C	Washer	456	Pressure limiting valve
319D	Spring	458	Repair kit
319E	Bolt	460	Pipe socket
319F	Retaining ring	464	Thrust ring
319M	Repair kit	466	Sealing ring
320	Repair kit, seals	468	Nut
320D	Repair kit, sleeves	510	Bracket
321	Bolt	560E	Bolt
325	Nut	752	Inlet union
330	Double adapter	756	Banjo bolt
335	Thrust ring	760	Sealing ring
336	Sealing ring	762	Pipe socket
336D	Nut	768	Inlet union
338	Banjo bolt		

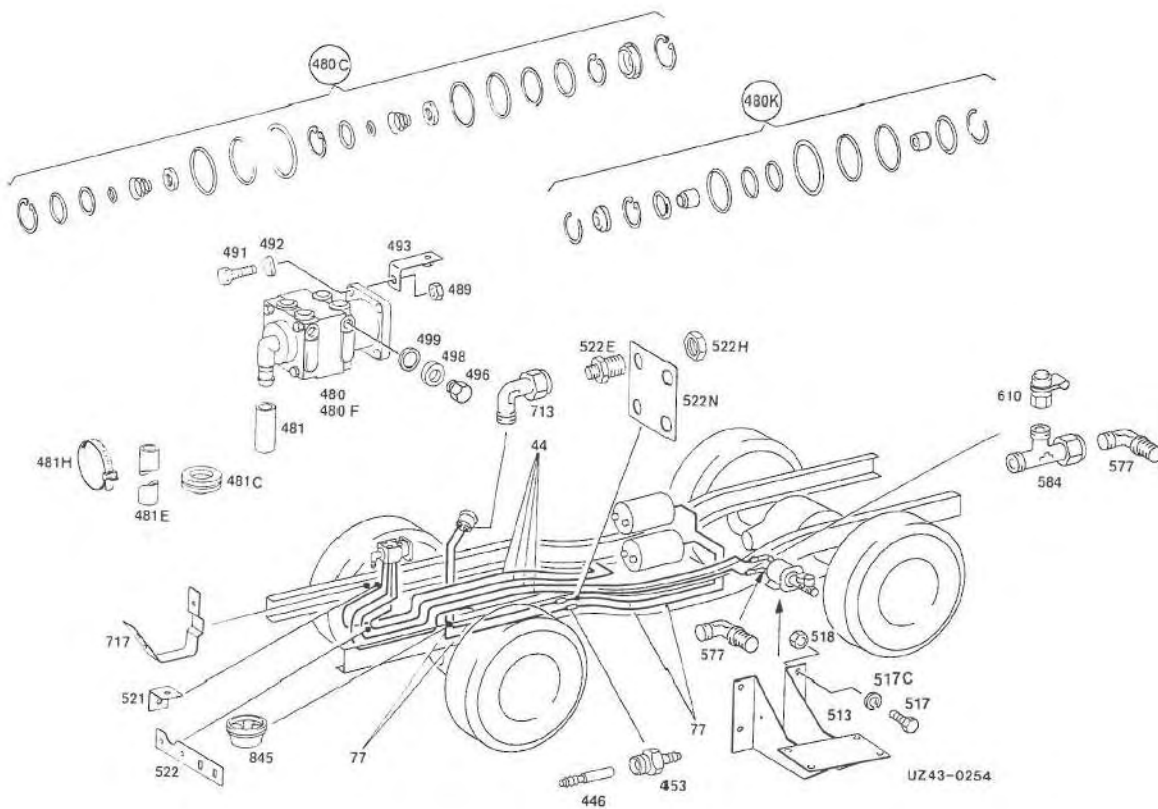
Exploded View



Air tanks

20	Pipe	398	Sealing ring
24	Bite ring	423	T-piece
25	Union nut	426	Sealing ring
29	Bite ring	428	Thrust ring
32	Union nut	430	Nut
35	Thrust ring	432	Pipe socket
38	Sealing ring	436	Test connection
41	Line	441	Bleeder valve
50	Union nut	444	Thrust ring
53	Bite ring	445	Sealing ring
69	Line	446	Pipe socket
77	Hose	450	Pipe socket
86	Union nut	704	Pipe
89	Bite ring	708	Hose
374	Air tank	710	Brake hose
377	Bracket	715	Bracket
382	Turnbuckle nut	719	Bolt
384	Nut	722	Nut
387	Mount	812	Clip
392	Bolt	814	Bolt
394	Nut	817	Lock washer
396	Screw plug	820	Nut

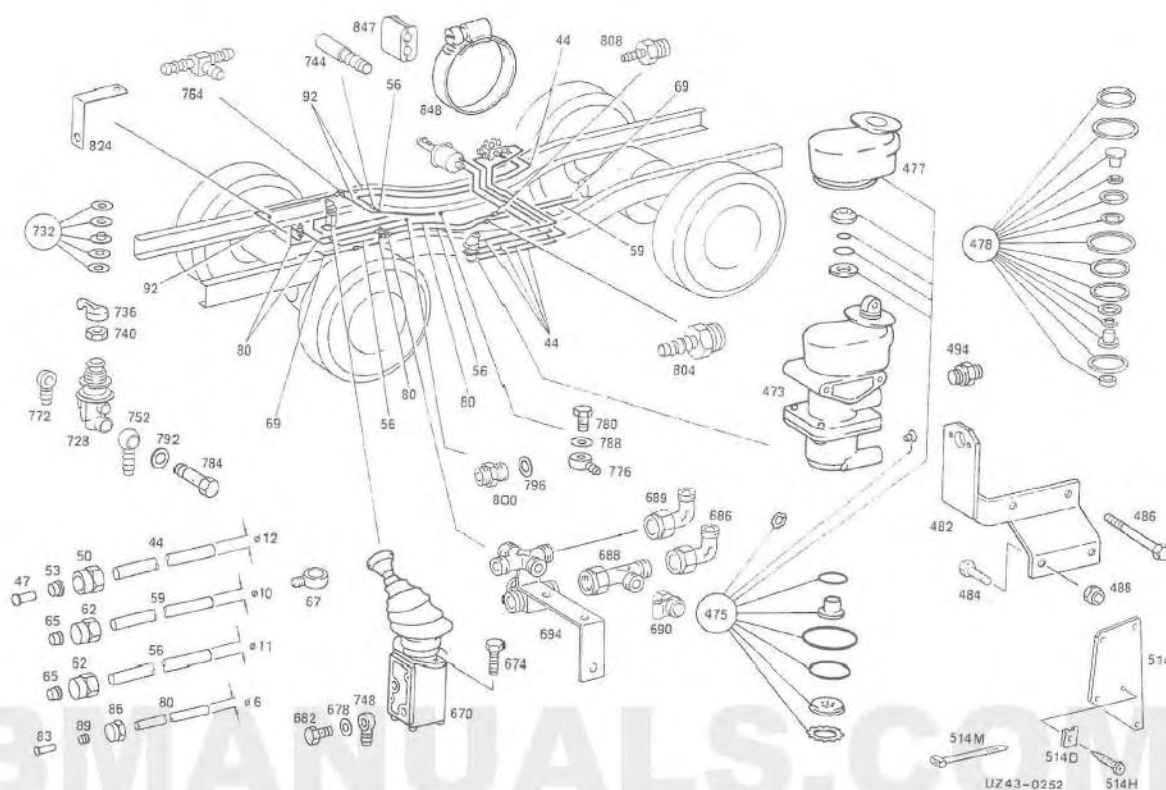
Exploded View



Brake System

44	Hose	498	Thrust ring
77	Hose	499	Sealing ring
446	Pipe socket	513	Bracket
453	Pipe socket	517	Bolt
480	Brake valve tractor, optional	517C	Lock washer
481C	Repair kit	518	Nut
480F	Brake valve tractor, optional	521	Bracket
480K	Repair kit	522	Bracket
481	Hose	522E	Pipe socket
481C	Sleeve	522H	Nut
481E	Rubber sleeve	522N	Shim
481H	Clip	577	Pipe elbow
489	Nut	584	Pipe socket
491	Bolt	610	Test connection
492	Spring washer	713	Pipe socket
493	Bracket	714	Bracket
496	Screw plug	845	Sleeve

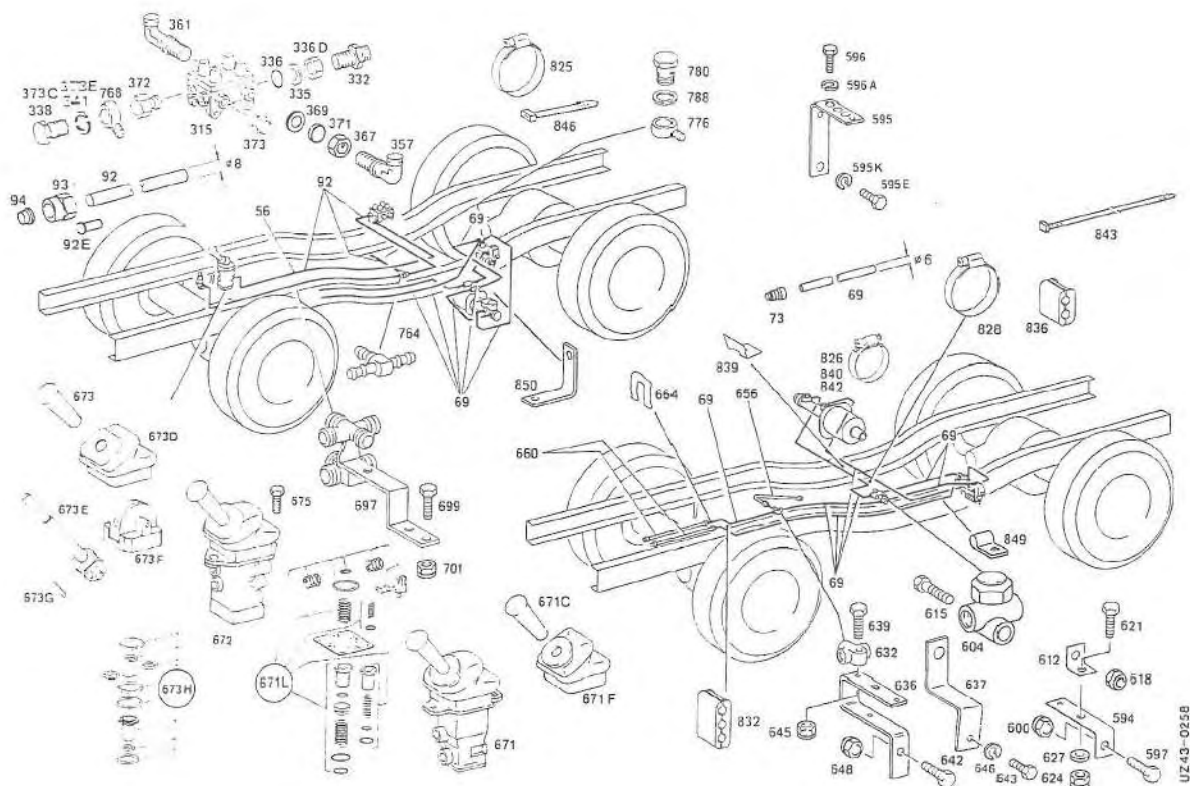
Exploded View



Brake system

44	Hose	514H	Screw	796	Sealing ring
47	Sleeve	514M	Clip	800	Pipe socket
50	Union nut	670	Parking brake valve	804	Pipe socket
53	Bite ring	674	Bolt	808	Pipe socket
56	Hose	678	Sealing ring	824	Bracket
59	Line	682	Banjo bolt	847	Mount
62	Union nut	686	Pipe socket	848	Clip
65	Bite ring	688	Pipe socket		
67	Line	689	Pipe socket		
69	Line	690	Test connection		
80	Hose	694	Bracket		
83	Sleeve	728	Compressed air valve – differential lock		
86	Union nut	732	Repair kit		
89	Bite ring	736	Knob		
92	Hose	740	Nut		
473	Brake valve	744	Pipe socket		
475	Repair kit	748	Inlet union		
477	Dust cap	752	Inlet union		
478	Repair kit	764	Connection piece		
482	Bracket	772	Inlet union		
484	Bolt	776	Inlet union		
486	Bolt	780	Banjo bolt		
488	Nut	784	Banjo bolt		
494	Screw union	788	Sealing ring		
514	Bracket	792	Sealing ring		
514D	Retainer				

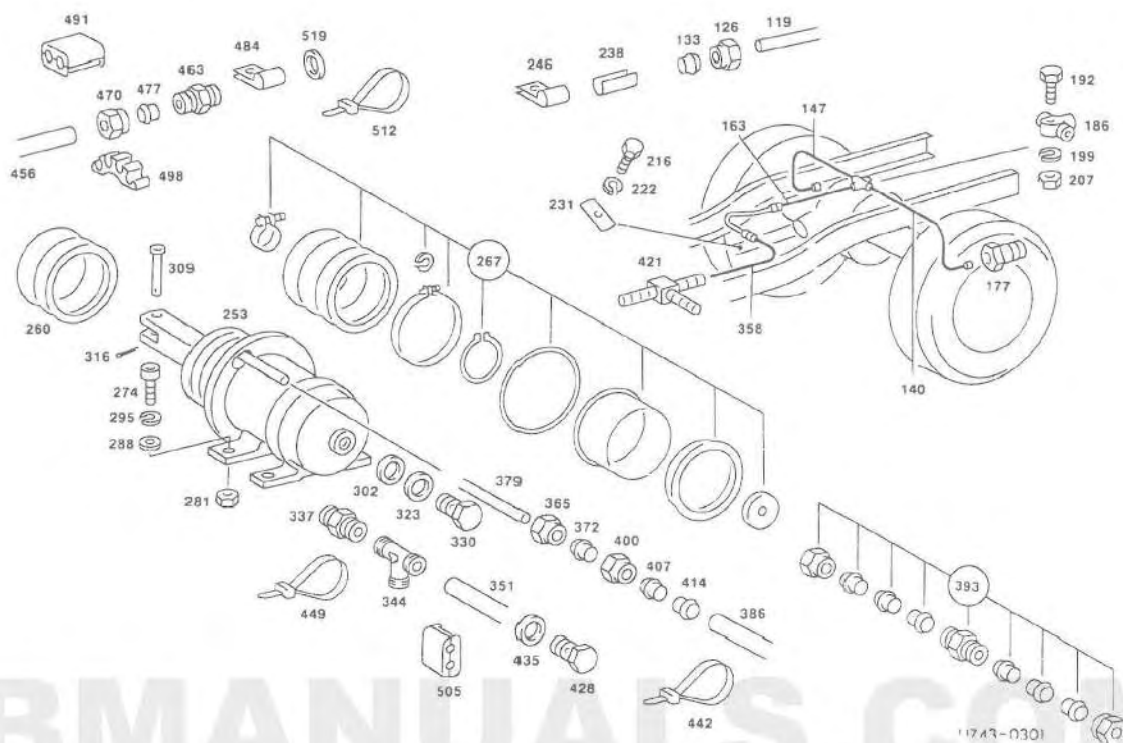
Exploded View



Brake system

56	Hose	596A	Lock washer	673	Grip
69	Line	597	Bolt	673D	Sleeve
73	Union screw	600	Nut	673E	Actuating unit
92	Hose	604	Test connection	673F	Cover
92E	Sleeve	612	Bracket	673G	Pin
93	Union nut	615	Bolt	673H	Repair kit
94	Bite ring	618	Nut	675	Bolt
315	Four-circuit protection valve	621	Bolt	697	Bracket
332	Pipe socket	624	Nut	699	Bolt
335	Thrust ring	627	Washer	701	Nut
336	Sealing ring	632	Distributor	764	Connection piece
336D	Nut	636	Bracket	768	Inlet union
338	Banjo bolt	637	Bracket	776	Inlet union
341	Sealing ring	639	Bolt	780	Banjo bolt
357	Pipe socket	642	Bolt	788	Sealing ring
361	Pipe socket	643	Bolt	825	Clip
367	Nut	645	Nut	826	Clip
369	Sealing ring	646	Lock washer	828	Clip
371	Thrust ring	648	Nut	832	Mount
372	Threaded piece	656	Brake hose	836	Mount
373	Screw union	660	Brake hose	839	Bracket
373C	Banjo bolt	664	Retainer	840	Clip
373E	Sealing ring	671	Parking brake valve, optional	842	Clip
594	Bracket	671C	Grip	843	Clip
595	Bracket	671E	Sleeve	846	Clip
595E	Bolt	671L	Repair kit	849	Clip
595K	Lock washer	672	Parking brake valve, optional	850	Bracket
596	Bolt				

Exploded View



Spring-Loaded Parking Brake

- 60 Spring-loaded cylinder
- 64 Repair kit for spring-loaded cylinder
- 69 Sleeve
- 108 Repair kit for line breakage

SA 36 043

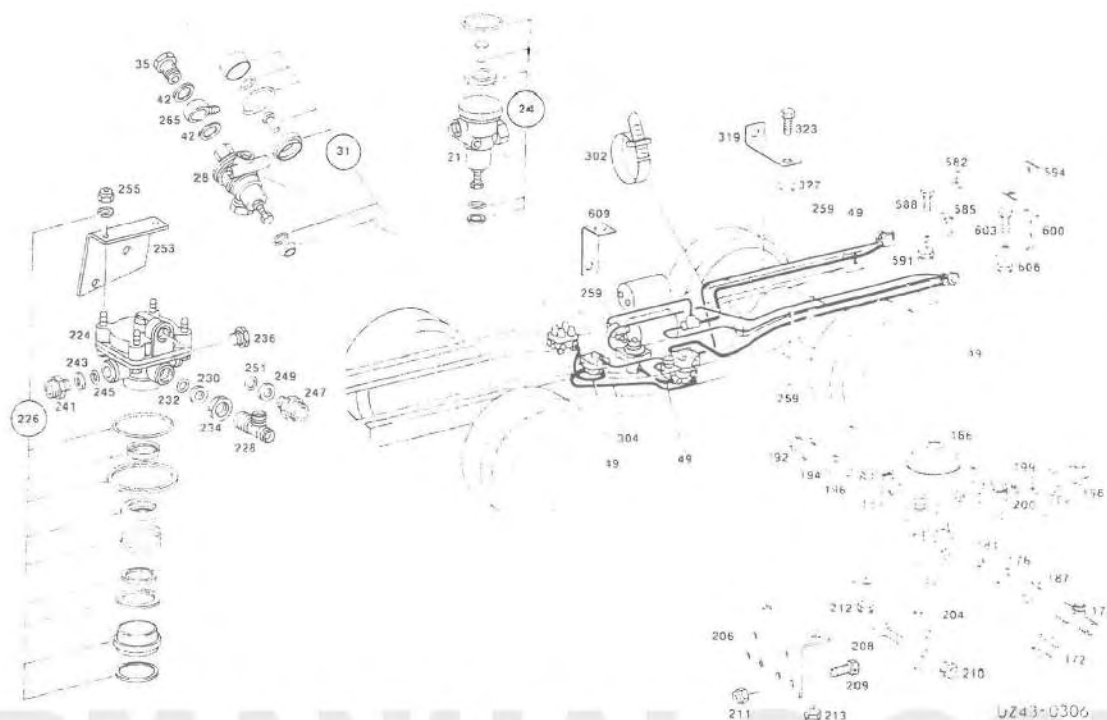
This diagram illustrates the assembly of a vehicle chassis and suspension system. The central component is the chassis frame, which includes the front and rear axles. The front suspension system is shown with coil springs, shock absorbers, and control arms. The rear suspension system is depicted with a leaf spring and a shock absorber. Various components are labeled with part numbers, including wheels, tires, axles, springs, suspension arms, and fasteners. The diagram is a technical drawing showing the exploded view of the components, allowing for identification of individual parts and their assembly sequence.

Key components and their part numbers include:

- Wheels and Tires: 407, 409, 412, 414, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918, 919, 920, 921, 922, 923, 924, 925, 926, 927, 928, 929, 930, 931, 932, 933, 934, 935, 936, 937, 938, 939, 940, 941, 942, 943, 944, 945, 946, 947, 948, 949, 950, 951, 952, 953, 954, 955, 956, 957, 958, 959, 960, 961, 962, 963, 964, 965, 966, 967, 968, 969, 970, 971, 972, 973, 974, 975, 976, 977, 978, 979, 980, 981, 982, 983, 984, 985, 986, 987, 988, 989, 990, 991, 992, 993, 994, 995, 996, 997, 998, 999, 1000.

53	Repair kit for line breakage	379	Bracket
262	Repair kit for line breakage	381	Bracket
305	Repair kit for line breakage	394	Coupling head, brake
350	Coupling head, supply	395	Coupling head, brake
351	Coupling head, supply	396	Sealing ring
352	Sealing ring	397	Sealing ring
353	Cover	398	Repair kit for coupling head
354	Repair kit for coupling head	407	Pipe socket
356	Repair kit for coupling head	414	Coupling head single-line
357	Pipe socket	431	Pipe socket
374	Bracket		

Exploded View



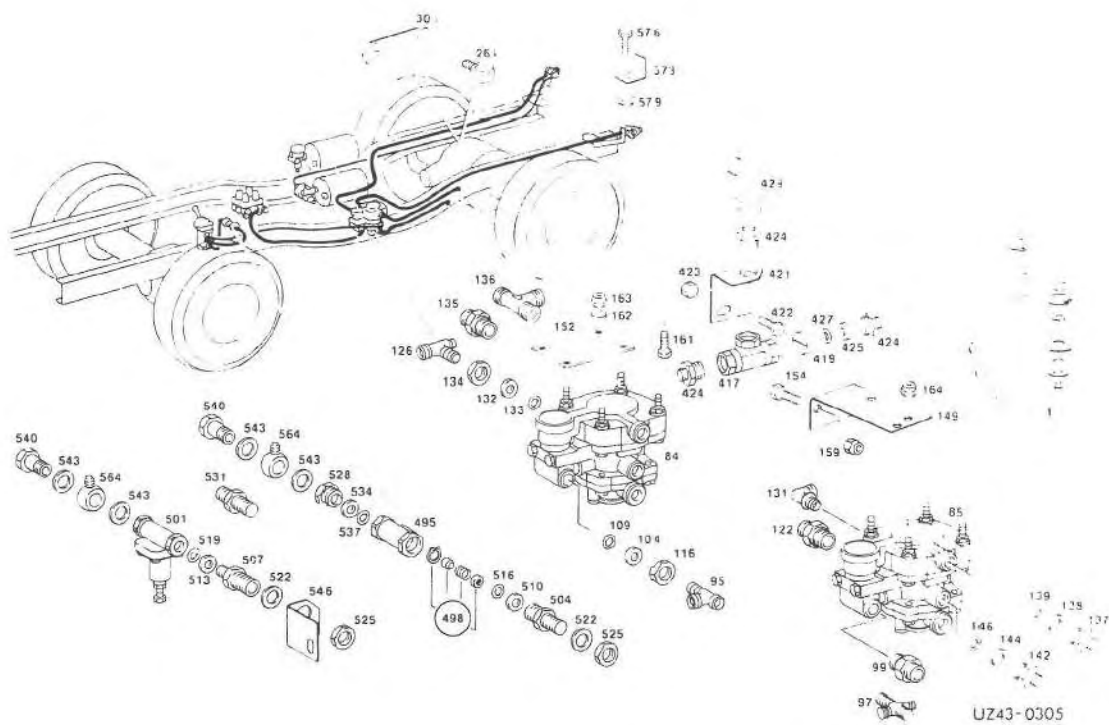
Trailer Brake System

- 21 Pressure limiting valve
- 24 Repair kit for pressure limiting valve
- 28 Pressure limiting valve
- 31 Repair kit for pressure limiting valve
- 166 Trailer control valve single-line
- 224 Relay valve
- 226 Repair kit for relay valve

43.14 General

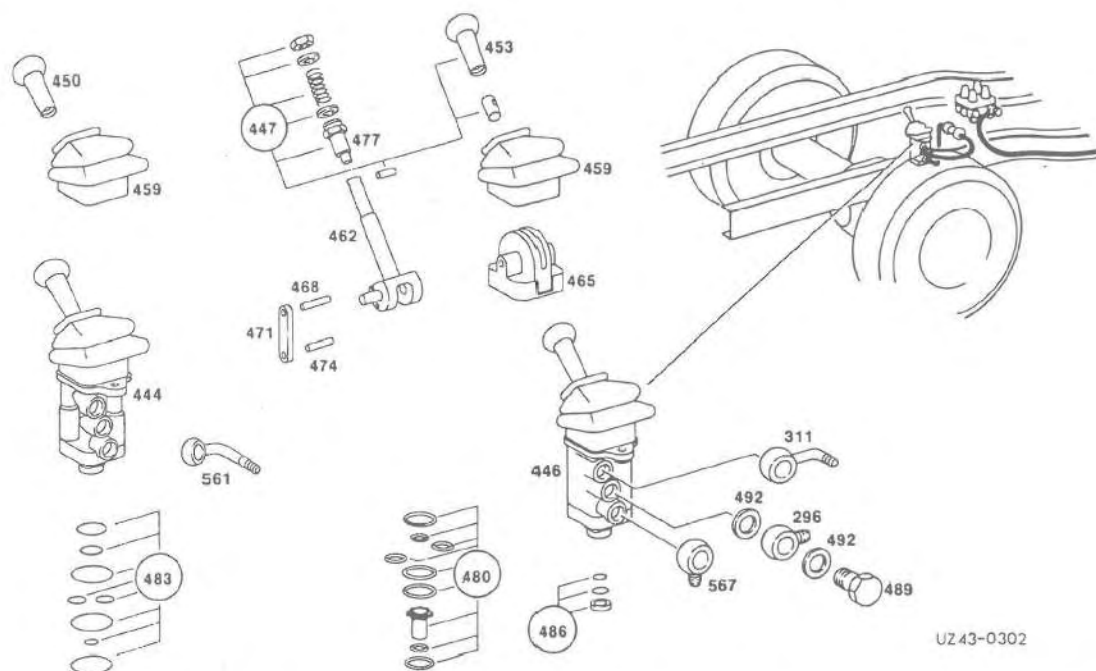
SA 36 043

Exploded View



Trailer Brake System

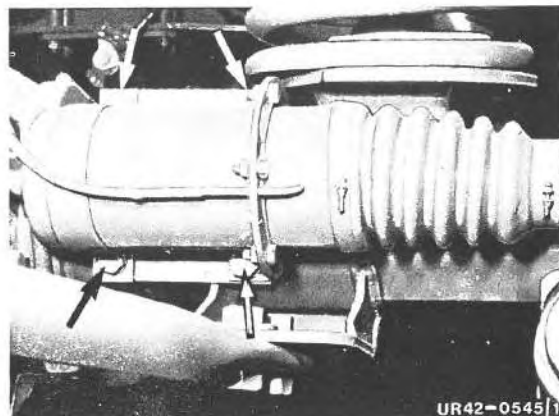
- 84 Trailer control valve 2-line
- 85 Trailer control valve 2-line
- 86 Repair kit for trailer control valve
- 417 Two-way valve
- 495 Non-return valve
- 498 Repair kit for non-return valve
- 501 Non-return valve with limited return flow

Exploded View**Trailer Brake System**

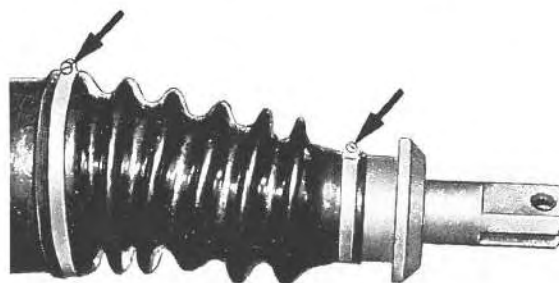
- 444 Parking brake valve
- 446 Parking brake valve
- 447 Lever
- 459 Sleeve
- 465 Cover
- 480 Repair kit for parking brake valve
- 483 Repair kit for parking brake valve

Renewing

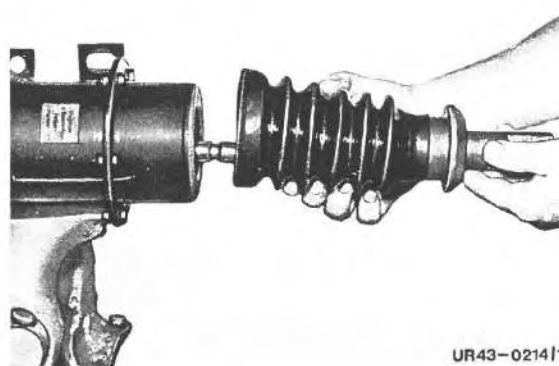
- 1 Remove wheel.
- 2 Release spring-loaded cylinder and remove.



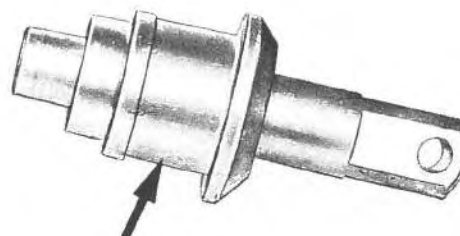
- 3 Detach hose clips.



- 4 Remove emergency release mechanism.

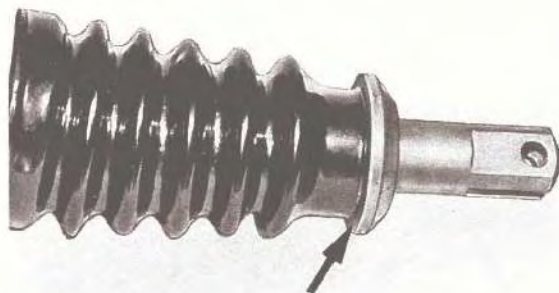


- 5 Clean contact surface for bellows on emergency release mechanism.



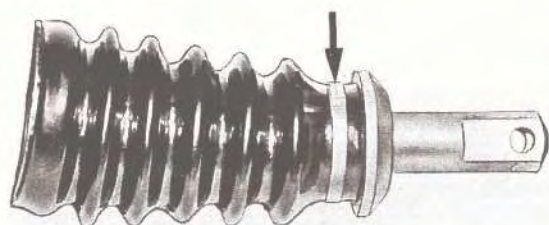
43.14 Renewing Bellows on Spring-Loaded Cylinder

SA 35 966



UR43-0216|1

6 Slide on new bellows up to the collar on the emergency release mechanism.



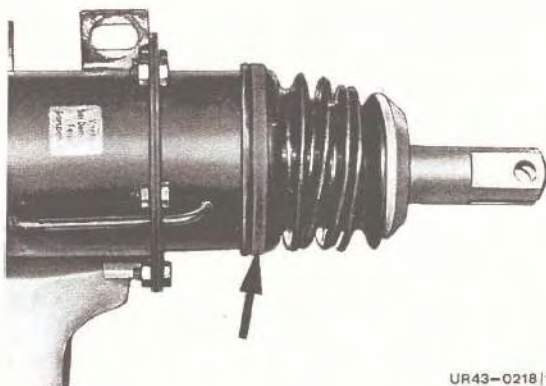
UR43-0217|1

7 Secure bellows on emergency release mechanism.

Note: New **clips** should be used for WABCO spring-loaded cylinders.

The old **hose clips with worm screw** can be used for GRAU spring-loaded cylinders.

8 Coat locking balls with water pump grease and insert in emergency release mechanism.

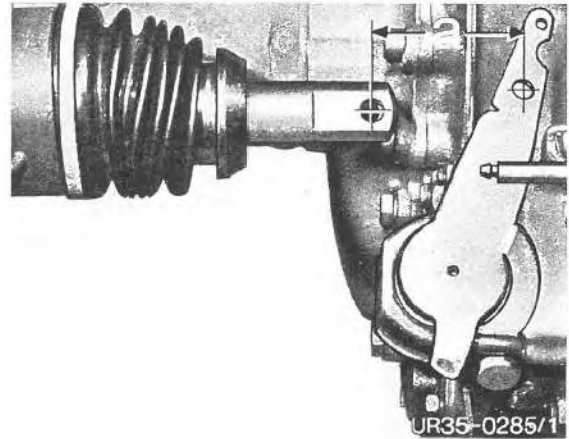


UR43-0218|1

9 Slide emergency release mechanism until it engages on the piston rod of the spring-loaded cylinder and secure bellows with hose clip at cylinder.

Note: Use new **clips** for WABCO spring-loaded cylinder. The old **hose clips with worm screw** can be used for GRAU spring-loaded cylinders.

- 10 Install spring-loaded cylinder.
- 11 Adjust spring-loaded cylinder relevant to fixed caliper.
See 35.3-3.1/5.
- 12 Pressurize spring-loaded cylinder and attach to actuating lever.
- 13 Fit wheel.



Survey

Version

Worm and nut power steering LS 3 B

46.11

Worm and nut power steering LS 7 F

46.12

Power steering pump

ZF vane pump ZF 7673

46.23

ZF vane pump ZF 7672

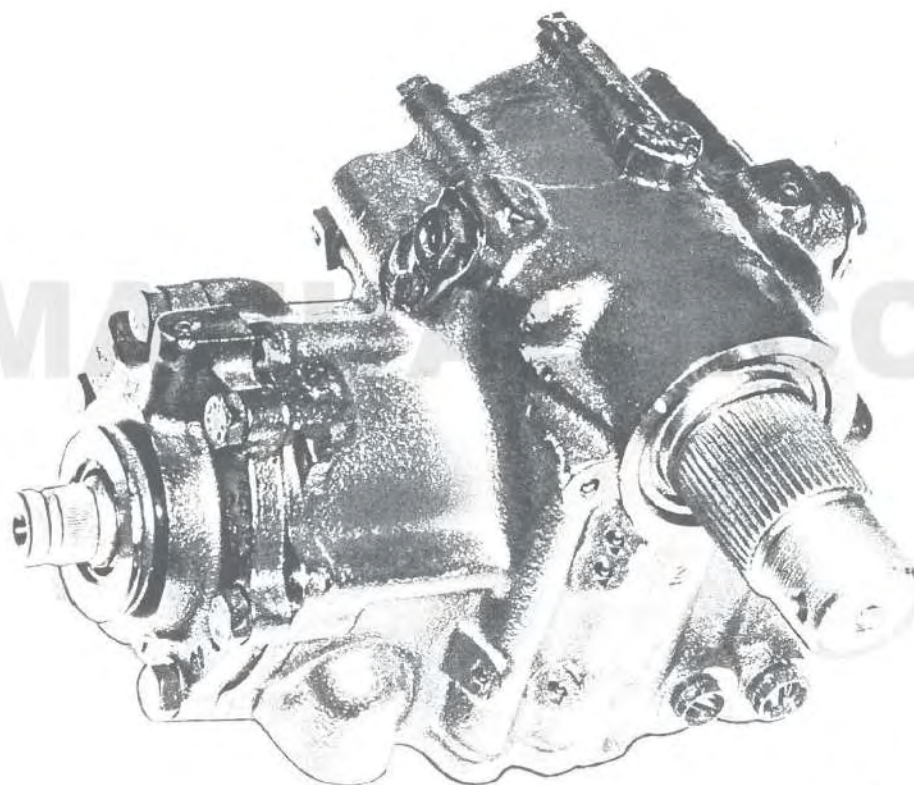
46.24

Contents

Chapter	Page
1 General	
Installation survey, steering	1.1/1
General view	1.1/1
Technical data, steering gear	1.2/1
Technical data, steering wheel	1.2/1
Adjustments	1.2/1
Capacities	1.2/1
Steering designation	1.2/2
Special tools	1.2/2
Necessary material	1.2/2
Tightening torques	1.3/1
Installation survey, pitman arm	1.4/1
General view	1.4/1
Technical data	1.4/1
Exploded view	1.5/1
2 Removal and installation of steering box	2.1/1
3 Disassembly and assembly of steering box	—
4 Adjustment of steering box	
Checking and adjustment of friction torque	4.1/1
Adjustment of steering limiter	4.2/1
5 Checking and adjustment of wheel lock	5.1/1
6 Checking and adjustment of wheel alignment	6.1/1
7 Checking tightness of universal joint at steering box	7.1/1

Installation survey

Chassis Model	Sales designation	Steering Model	Sales designation	Manufacturer	Type	Installation
435.110	U 1700	765.601	LS 3 B	Daimler-Benz	Worm and nut power steering	Standard
.111	U 1700 L					
.113	U 1700 L/38					
.115	U 1300 L					
.117	U 1300 L/37					



UR 46-0165

Steering box LS 3 B

Technical data**Steering box**

Ratio in steering box		19.33 : 1
Worm pitch, left-hand	mm	13
Piston diameter	mm	107
Swept piston area	cm ²	81.37
Max. rotation of steering shaft	°	90
Spline module	mm	8
Pitch radius	mm	40
Piston stroke	mm	2 x 31.5
Hydraulic torque at 100 bar	Nm	3255

Steering wheel

Steering wheel diameter	mm	450
Permissible play at steering wheel circumference, wheel at center position and engine running	mm	40
Steering wheel turns from lock to lock		4.5

Adjustments

Distance (A) from side member to center of hole in pitman arm	mm	165
Fiction torques of steering worm at center position	Ncm	200
	Ncm	250
Wheel alignment (toe-out)	mm	0 to -4

Capacities

Steering Model designation	Sales designation	Service products ¹⁾	Quantity L
765.601	LS 3 B	Engine oil SAE 10 W ²⁾	2.25
		ATF	
		Hydraulic fluid ³⁾	

¹⁾ Oils listed must not be mixed.

²⁾ Use SAE 5 W - 20/30 in cold zones.

³⁾ Valid for military vehicles

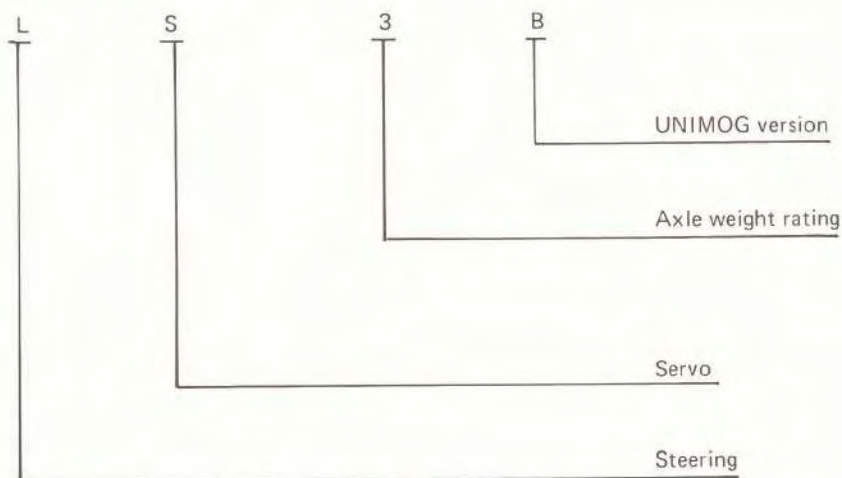
a) Do not use fluids other than those approved by Daimler-Benz.

b) Use approved fluids in accordance with vehicle operator's instructions.

In the absence of instructions from vehicle operator, use the regular approved fluids.

765.601

Steering designation



Special tools

Cons.No.	Tool	Number	Tool set
1	Puller for steering wheel	355 589 00 33 00	B
2	Puller for drag link and track rod	406 589 05 33 00	B
3	Puller for pitman arm	406 589 06 33 00	B
4	Puller for steering wheel, with tool No. 1	425 589 01 34 00	B
5	Torque meter for friction torque of steering worm	001 589 49 21 00	C
6	Torque meter to adjust limiter	000 589 67 21 00	C
7	Handwheel for friction torque of steering worm, with tool No. 5	352 589 08 63 00	C

Necessary materials

Cons.No.	Designation	Part No.
1	Longlife lubricating grease	regular

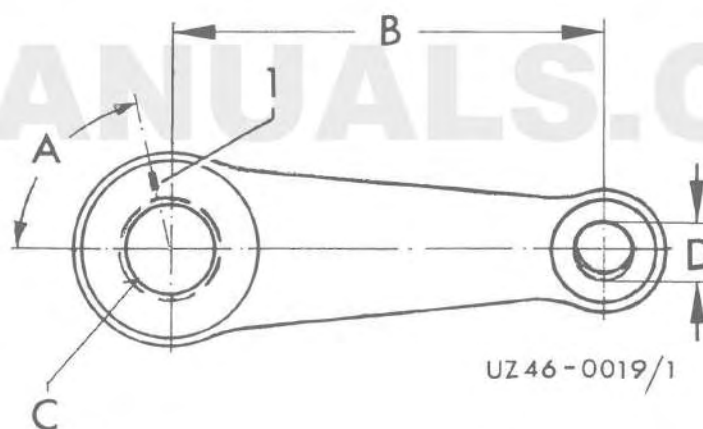
Tightening torques

Location	Thread	Nm
Steering box to frame	M 18 x 1.5 x 65	420
Cover to steering box	M 10 x 25	73 to 78
Adjusting screw of steering shaft	M 12	60 to 70
Retaining bolts for limiter	M 6 x 12	8 to 10
Pitman arm to steering shaft	M 42 x 1.5	530
Steering knuckle arm and steering arm to steering knuckle	M 20 x 1.5	450
Track rod to steering knuckle arm and steering arm	M 20 x 1.5	230 ± 20
Drag link to pitman arm and steering knuckle arm	M 20 x 1.5	230 ± 20
Clamp to track rod	M 12 x 1.5 x 55	90 ± 3
Clamp to drag link	M 12 x 1.5 x 50	90 ± 3
Steering shaft to steering	M 10 x 45	64
Steering wheel to steering shaft	BM 22 x 1.5	50
Steering joint mounting	M 8 x 42	33

Installation survey

Chassis Model	Sales designation	Steering Sales designation	End chassis No.	Pitman arm MB part No.	Installation
435.110	U 1700	LS 3 B	as of beginning of production	425 463 04 01	Standard
.111	U 1700 L				
.115	U 1300 L				
.117	U 1300 L/37				

General view

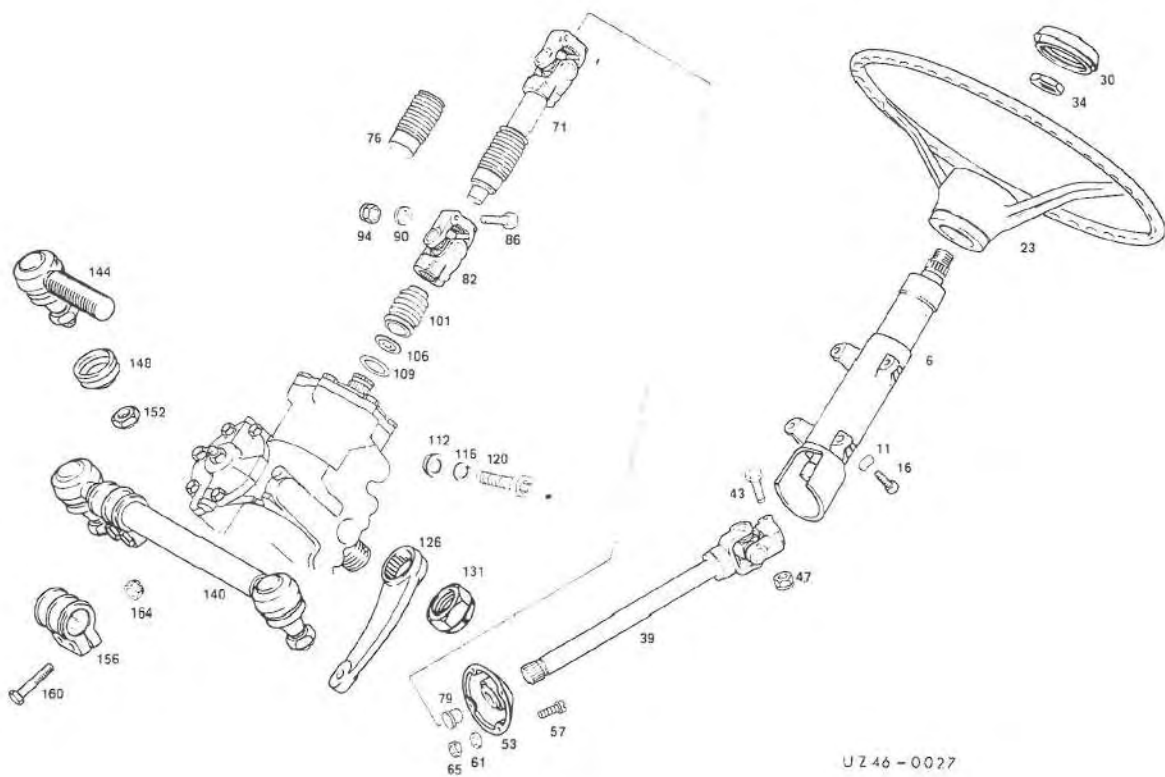


Pitman arm

Technical data

Mark at center of tooth gap	A	degrees	$80 \pm 30'$
Hole spacing	B	mm	205
Tapered splining N 446	C	"	2
Taper 1:10	D	mm	dia. 26 H 8

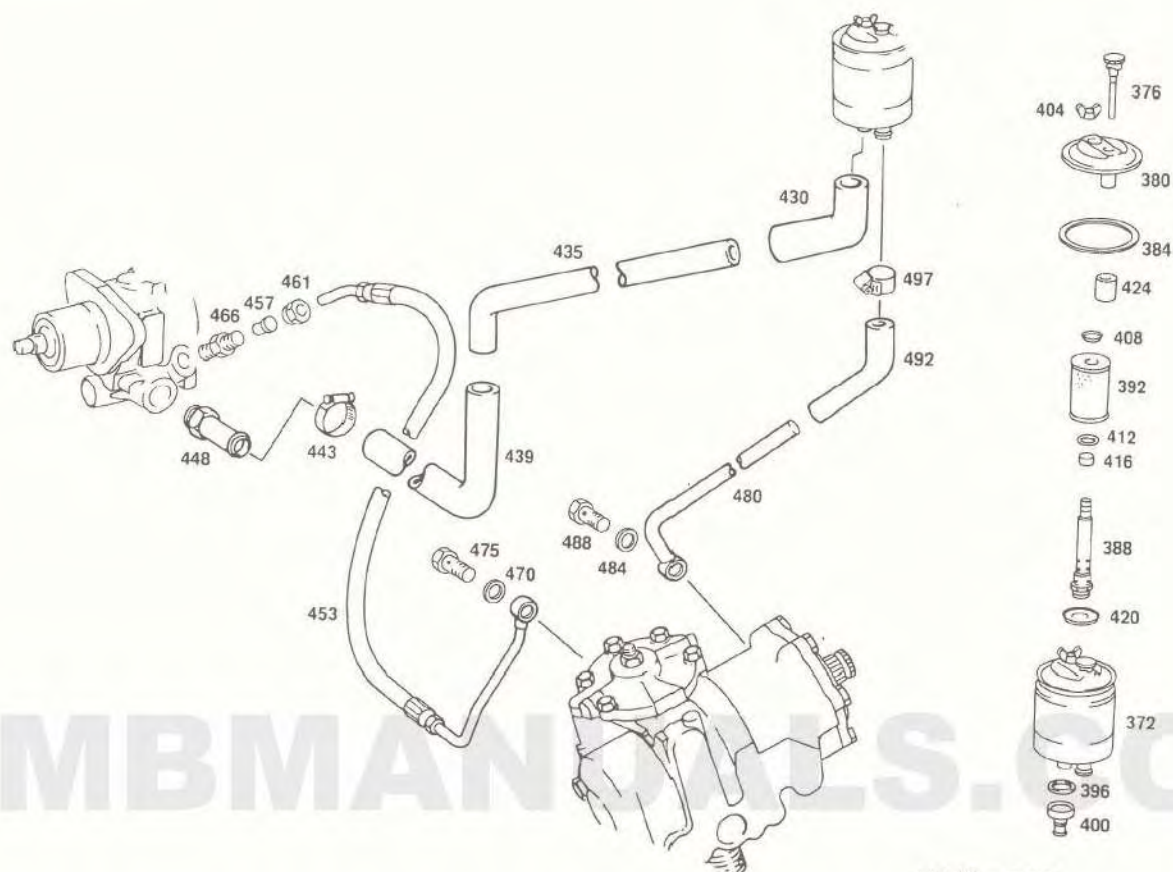
Exploded view



Steering wheel with steering linkage

6	Steering column bracket	94	Nut
11	Spring washer	101	Gaiter
16	Bolt	106	Washer
23	Steering wheel	109	Washer
30	Cap	112	Spherical washer
34	Nut	116	Lock washer
39	Steering shaft	120	Bolt
43	Bolt	126	Pitman arm
53	Bearing	131	Nut
57	Bolt	140	Drag link
61	Washer	144	Joint
65	Nut	148	Gaiter
71	Steering shaft	152	Nut
76	Gaiter	156	Clamp
82	Universal joint	160	Bolt
86	Bolt	164	Nut
90	Washer		

Exploded view

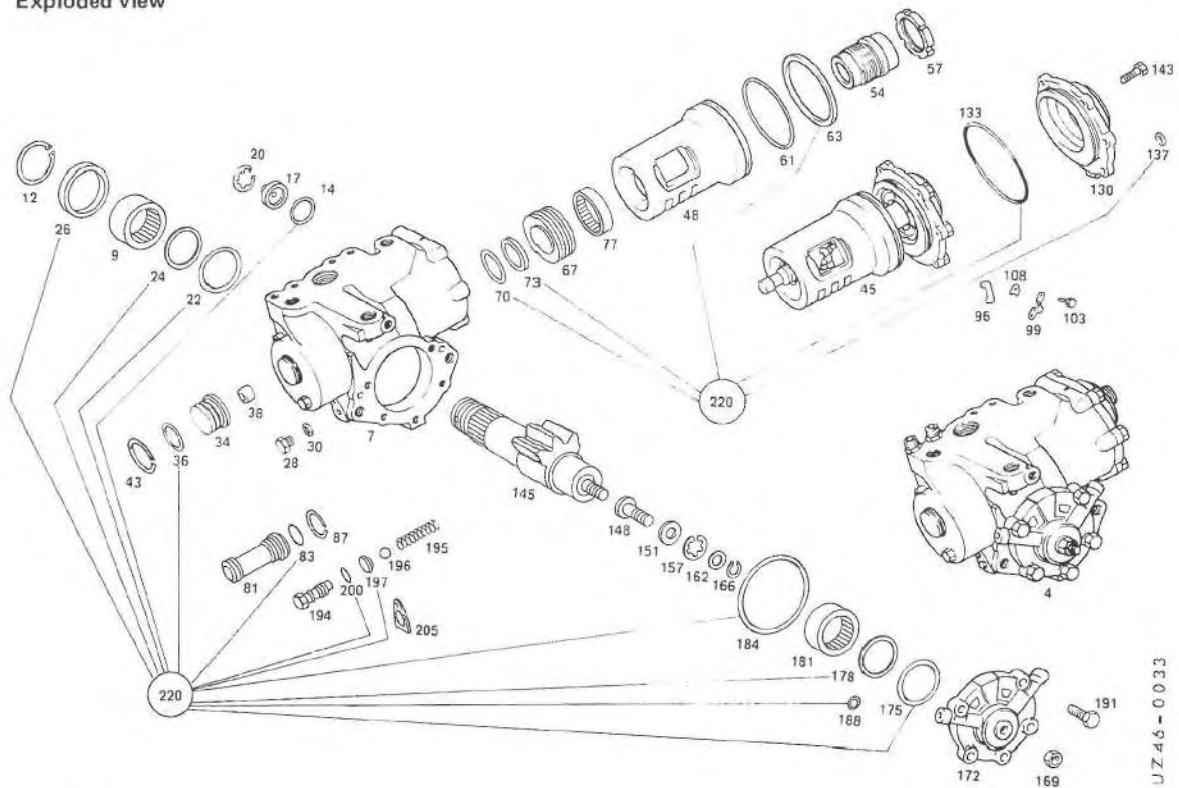


UZ46-0031

Oil reservoir with connecting pipes

372	Oil reservoir	435	Line
376	Pipe	439	Rubber sleeve
380	Cap	443	Clamp
384	Sealing ring	448	Pipeline
388	Filter support	453	Line
392	Filter	457	Bite ring
396	Sealing ring	461	Cap nut
400	Union	466	Screwed union
404	Nut	470	Sealing ring
408	Washer	475	Union screw
412	Washer	480	Pipeline
416	Spacing tube	484	Sealing ring
420	Cover	488	Union screw
424	Rubber sleeve	492	Rubber sleeve
430	Rubber sleeve	497	Clamp

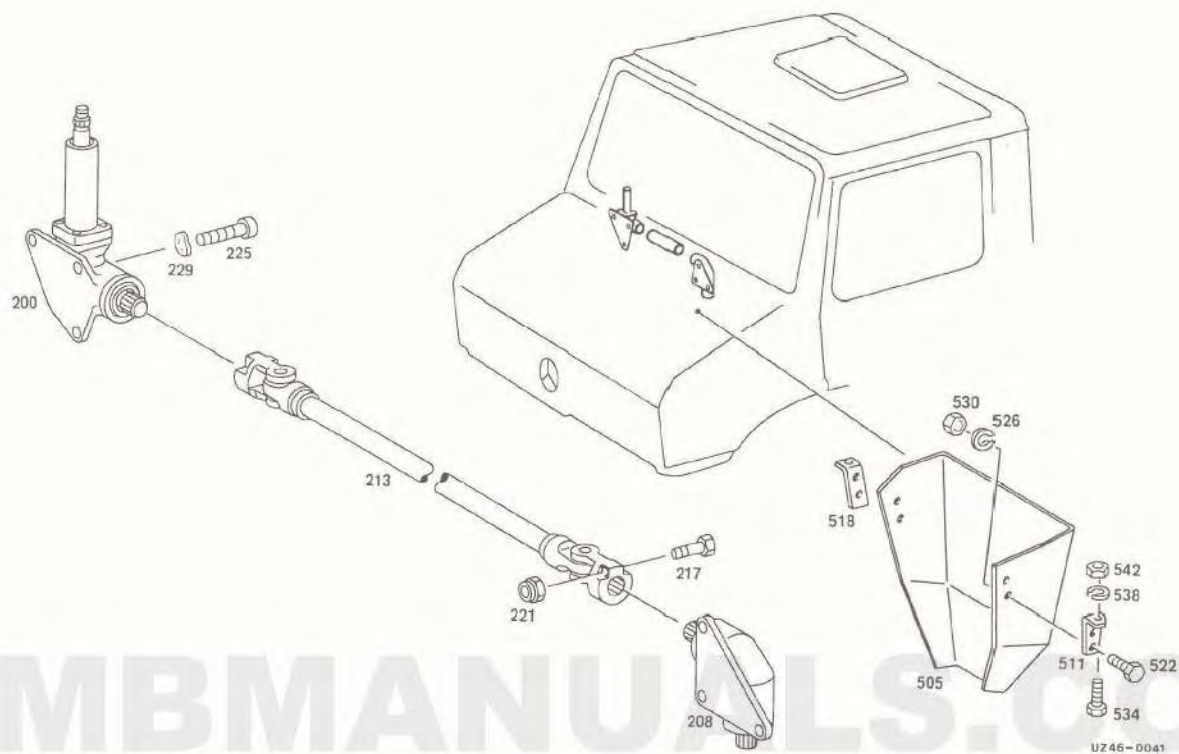
Exploded view



Steering box

4	Steering	96	Ball guide
7	Housing	99	Clip
9	Needle bearing	103	Bolt
12	Retainer	108	Snap ring
14	Sealing ring	130	Cover
17	Cover	133	Sealing ring
20	Retainer	137	Sealing ring
22	Sealing ring	143	Bolt
24	Sealing ring	145	Gearing shaft
26	Sealing ring	148	Bolt
28	Screw plug	151	Thrust washer
30	Sealing ring	157	Snap ring
34	Bushing	162	Thrust washer
36	Sealing ring	166	Snap ring
38	Needle bearing	169	Nut
43	Snap ring	172	Cover
45	Piston	175	Sealing ring
48	Piston	178	Sealing ring
54	Insert	181	Needle bearing
57	Nut	184	Sealing ring
61	Sealing ring	188	Sealing ring
63	Sealing ring	191	Bolt
67	Screw collar	194	Valve
70	Sealing ring	195	Spring
73	Sealing ring	196	Ball
77	Needle bearing	197	Sealing ring
81	Bushing	200	Sealing ring
83	Sealing ring	205	Retainer
87	Snap ring	220	Repair set

Exploded view



Steering linkage for righthand drive

200	Bevel drive right	505	Bracket
208	Bevel drive left	511	Bracket
210	Steering linkage	518	Bracket
217	Bolt	522	Bolt
221	Nut	526	Washer
225	Bolt	530	Nut
229	Washer	534	Bolt
		538	Washer
		542	Nut

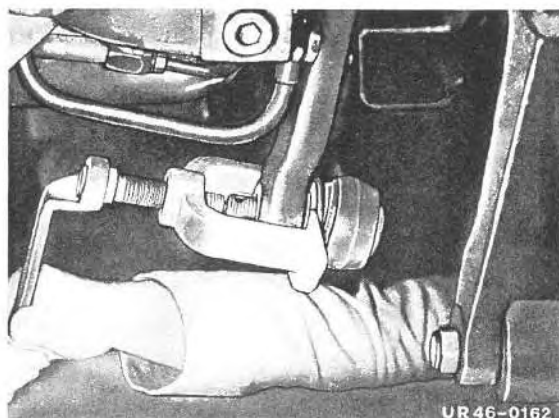
Removal

Note: Prior to removal, move wheels to straight-ahead position and check wheel alignment with reference to 5.1/1.

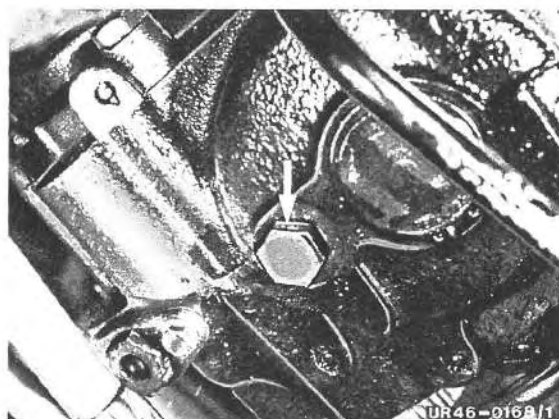
- 1 Remove engine hood and front panel
- 2 Unscrew castle nut and detach drag link using special tool No. 2.
- 3 Drain oil, moving pitman arm so that all oil runs out.
- 4 Unscrew delivery line (1) and return line (2).



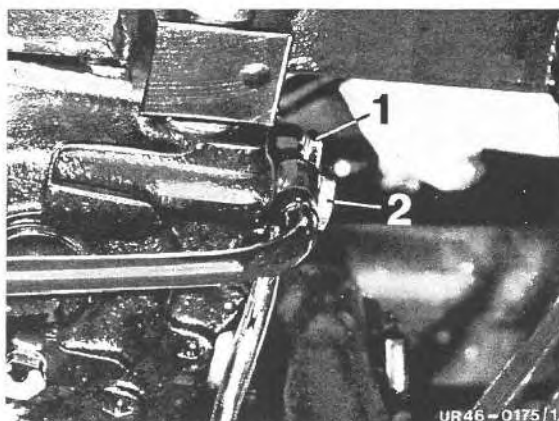
UR88-0030/3



UR46-0162

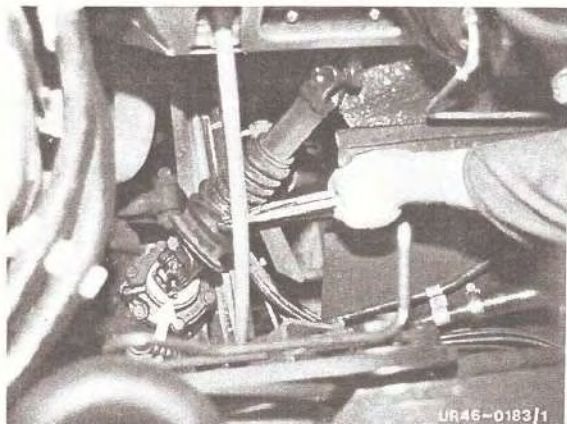


UR46-0168/1

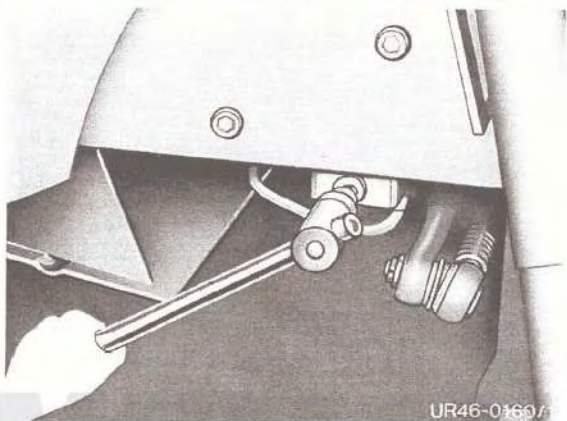


UR46-0175/1

765.601



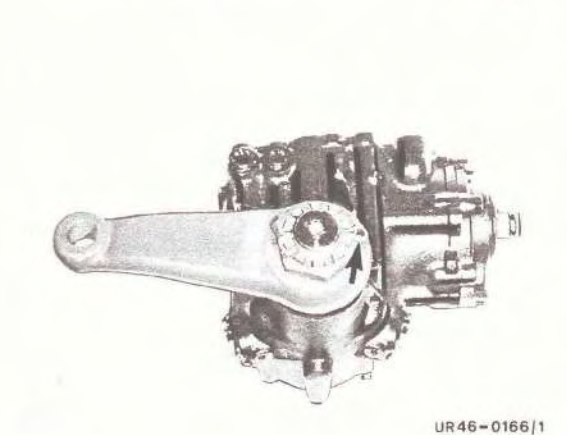
- 5 Detach universal joint at steering box.



- 6 Unscrew and remove steering box.



- 7 Release pitman arm and remove using special tool No. 3.



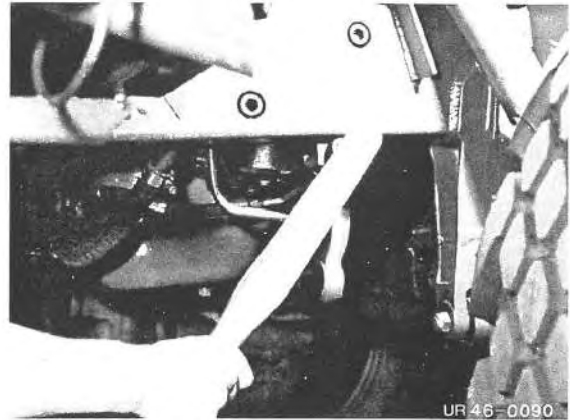
Installation

- 1 Attach pitman arm, torque and secure.
See 1.3/1 for tightening torques.

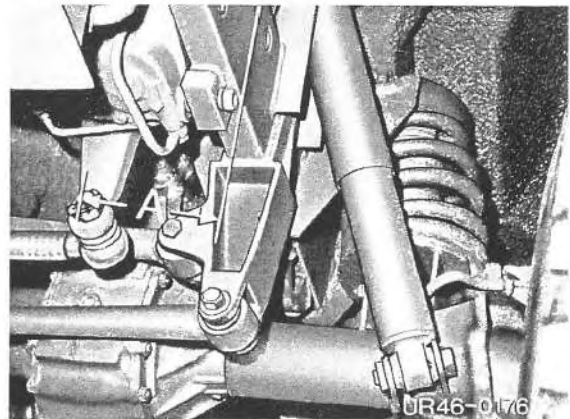
Note: Mark on pitman arm must point toward mark on steering shaft.

765.601

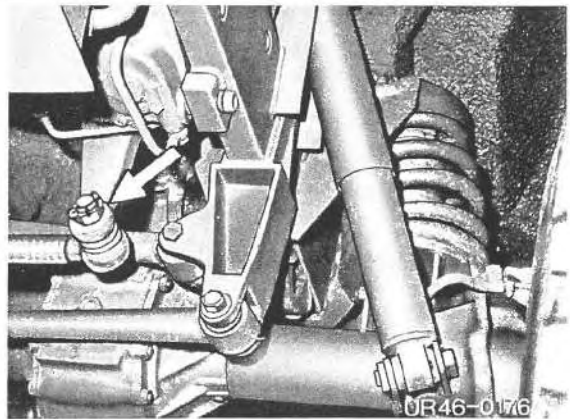
- 2 Attach steering box with conical washer and torque.
See 1.3/1 for tightening torques.



- 3 Attach drag link. Measure distance from side member to center of hole in pitman arm. If necessary, adjust drag link and track rod so that dimension (A) is obtained.
See 1.2/1 for dimension A.



- 4 Torque drag link and secure.
See 1.3/1 for tightening torques.



- 5 Attach delivery line (1) and return line (2).

